



Siddaganga Institute of Technology, Tumakuru-572 103

(An Autonomous Institution affiliated to VTU, Belagavi, Approved by AICTE, New Delhi)

Even Semester B.E. Information Sc. and Engg. Examinations July-Aug. 2024

Advanced Web Technology and Internet Applications

Time: 3 Hours

Max. Marks: 100

Note : 1. Revealing of Identity in any form in the answer book will be treated as malpractice.
2. Answer any five questions choosing one full question from each unit.

Unit - I

M BL CO PO PSO

- 1 a) Discuss three broad levels of web development. Differentiate between circuit switching and packet switching. Name any one of the application for these two switching techniques.
- b) What is the client-server model of communication? How does it differ from peer-to-peer?
- c) How many requests are involved in displaying a single web page? Describe.

8	2	1	1	2
6	2	1	1	2
6	2	1	1	2

OR

- 2 a) Describe the relative advantages and disadvantages of web-based applications in comparison to traditional desktop applications.
- b) How many levels can a domain name have? What are genetic top-level domains?
- c) Describe the four layers in the four-layer network model.

8	2	1	1	2
6	2	1	1	2
6	2	1	1	2

Unit - II

- 3 a) Describe the difference between a relative and an absolute reference. When should each be used?
- b) How does the <figure> element differ from the <image> element? In what situations does it make sense to use or not use <figure>?
- c) What are class selectors? What are id selectors? Briefly discuss why would use one over the other.

8	1	2	1	2
6	3	2	3	2
6	2	2	2	2

OR

- 4 a) What are the three cascade principles used by browsers when style rules conflict? Briefly discuss each other.
- b) Create a HTML document having class called income, and make the background color #0ff. Create a class called expenses, and make the background color #f0f. Create a class called profit, and make the background color #f00. Throughout the document, any text that mentions income, expenses, or profit, attach the appropriate class to that piece of text.
- c) What are contextual selectors? Identify the four different contextual selectors.

8	2	2	2	2
6	3	2	2	2
6	3	2	2	2

Unit - III

- 5 a) What is JavaScript? What are its relative advantages and disadvantages?
- b) Differentiate between server side and client side development.
- c) Write an external JavaScript to compute the best of two test marks average among three tests.

8	2	3	2	2
6	1	3	2	2
6	3	3	3	2

OR

- 6 a) Discuss the advantages and disadvantages of client-side scripting. What is an anonymous function? What is a nested function? What are some of the reasons for using these two types of functions?

10	3	3	2	2
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- b) Write a PHP program to create a class STUDENT with the following specifications.

Data members: name, Roll number, Average marks

Member function ; Read(getters) and write (setters)

Use the above specification to read and print the information of 2 students.

10 3 3 3 2

Unit - IV

- 7 a) Explain \$_SERVER and \$_FILE superglobal arrays.

8 3 4 1 2

- b) Discuss different types of errors and exceptions.

6 2 4 2 2

- c) Describe the ways that can limit the types and size of file uploaded.

6 3 4 2 2

OR

- 8 a) Differentiate between error and exception. Discuss with an example how exceptions are handled in PHP.

8 3 4 1 2

- b) How do you read or write a file on the server from PHP? Give example.

6 3 4 2 2

- c) What are the different types of in-memory file functions? Discuss.

6 3 4 2 2

Unit - V

- 9 a) What does \$() shorthand stands for in jQuery? Explain any 3 jQuery form selectors. Write a jQuery selectors to get all the <p> that contain the word "hello".

10 3 5 2 2

- b) Imagine that you are asked to provide advice on implementing web services for a site. Discuss the merits and drawbacks of SOAP and REST based web services and for XML versus JSON as a REST data format.

10 3 5 3 2

OR

- 10 a) Discuss jQuery selectors in detail.

8 3 5 2 2

- b) What are the advantages of using asynchronous requests over traditional synchronous ones?

6 2 5 2 2

- c) What is well-formedness and validity in the context of XML? How do they differ?

6 2 5 2 2